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Report On

Basic Project SatoshiFlow

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Abstract

In this work we explore the Technical Analysis (TA) way of trading by devising and implementing a robust trading strategy for BTC-USD using Moving Average Crossover method. Here we also study the effect of changing window size and weights on the trading of BTC.

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1 Strategy & Logic

Throughout this work we primarily use Moving Average Crossover (MAC). The way this works is one runs 2 MAs simultaneously, one whose window is small its called as Fast MA and it averages over time near today (short term). And the other MA is Slow MA which has longer window as it averages over time far from today (long term).

Once a Fast MA cuts above the Slow MA, then its called as Golden cross and this signals for buy. And when a Fast MA cuts below the Slow MA, then its called as Death cross and this signals for sell.

The core logic behind this method relies on the fact that when a Fast MA cuts above the Slow MA then it implies that the prices of near past higher than the far past which is a clear signal to buy. Whereas a Fast MA cutting below the Slow MA implies the near past prices are lower than far past prices which is clear signal signal to sell.

There are 2 (opposing) issues associated with MAC. Firstly, is lag effect. Since, we're measuring the relative difference between near past (Fast MA) and far past (Slow MA). This takes a time to signal. Let's say a news strike which happens to be a bad news for company A. Therefore, its stock prices are definitely going to fall. For a person following news regularly will be able to quickly deduct the above piece of information from the news today and be able to make move. But a person using MAC will incur some initial loss until (depending on its sensitivity which is related to moving window size) few days later algorithms signals a sell. Secondly, the whipsaw effect. Some time a stock price might fall for a short period and increases back. And when your MAC is too sensitive to trigger a signal to sell and such signal is called as False Signal. This only incur a small amount of loss due to transaction fee.

Types of MAC on the basis of window size:-

- Short windows size - 5/20: Fast MA averages over past 5 prices and Slow MA averages over past 20 prices.
- Long windows size - 50/200: Fast MA averages over past 50 prices and Slow MA averages over past 200 prices.

Types of MAC on the basis of mode of averaging:-

- Simple Moving Average (SMA): SMA averages over past prices on equal weights.
- Weight Moving Average (WMA): WMA averages over past prices such that the prices closer to present gets weighted more to keep track of recent events.
- Exponential Moving Average (EMA): EMA is an extreme version of WMA where its present EMA is computed to be weights sum of present price and previous EMA. Put even higher emphasis on recent price.

2 Performance Metrics

We use the net Profit & Loss as the metric. In our work we assume if you put 1000 USD at the start of time series of trading then how much do you get back at the end of time series of trading. Its conveniently expressed in percentage.

$$PnL = \frac{(FinalPrice - InitialPrice)100}{InitialPrice}$$

3 Comparison of Strategies

Window Size comparison:-

Here we kept the mode of moving averaging to be SMA for the comparing between 5/20 and 50/200. We observe a net profit of 417.0133538026703% for 5/20 Window and a net profit of 373.656134602649% for 50/200. There are few reasons for 50/200 underperforming 5/20, firstly, for 50/200 the large window size is 200 which means first 199 datapoints in time series is gone which in comparison for net datapoint of 1577. This loss of more than 12.5% datapoints makes it pretty delayed entry into trading. Secondly, BTC market value suffers from high reactivity which requires least lag that 5/20 window offers.

Mode of averaging comparison:-

Here we kept the window size to be 5/20 while comparing the 3 modes of MA. Here's the result:

SMA: 417.0133538026703%

WMA: 401.5561650744154%

EMA: 364.8204418330586%

Observe that SMA has the highest net profit while EMA has the least net profit. As EMA works by putting high weight on yesterday so, due to high reactive nature of BTC, EMA strategy suffers from sharp increase or decrease against the trend leading to loss which is popularly known as Whipsaw effect.

In conclusion, SMA 5/20 works best for this problem of trading BTC-USD.